COMPUTER ENGINEERING

QUEZON CITY

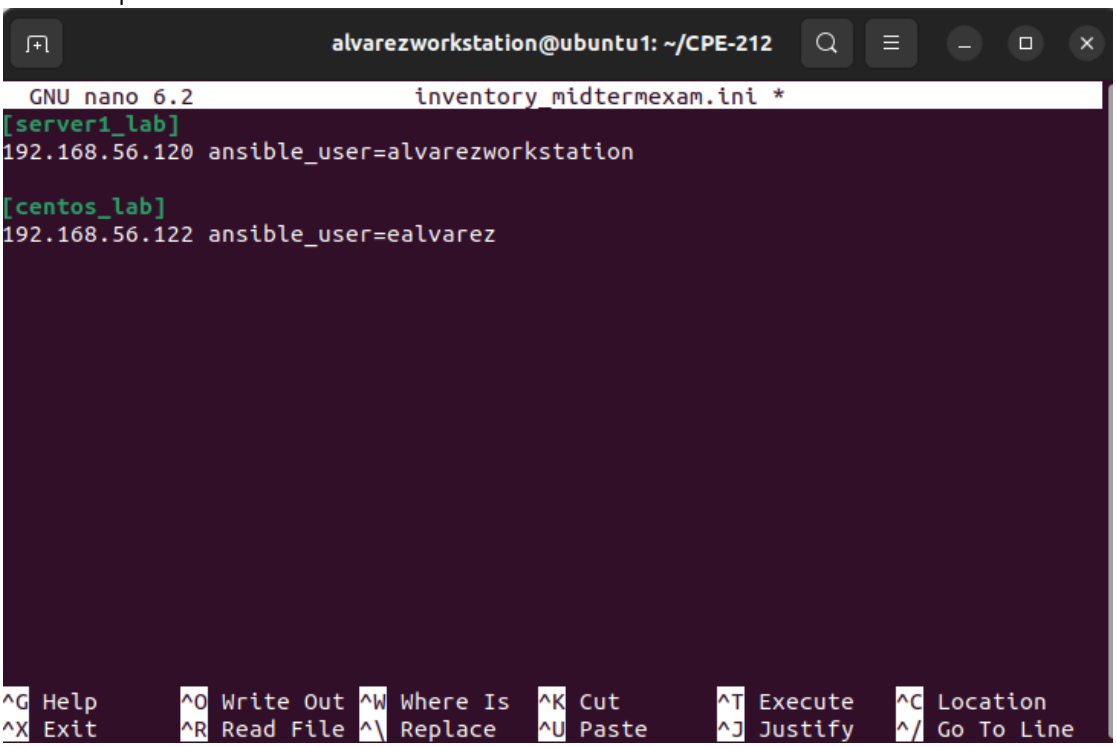
COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	09/24/2026	**DATE SUBMITTED:**	09/24/26
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Midterm **Skill Exam**

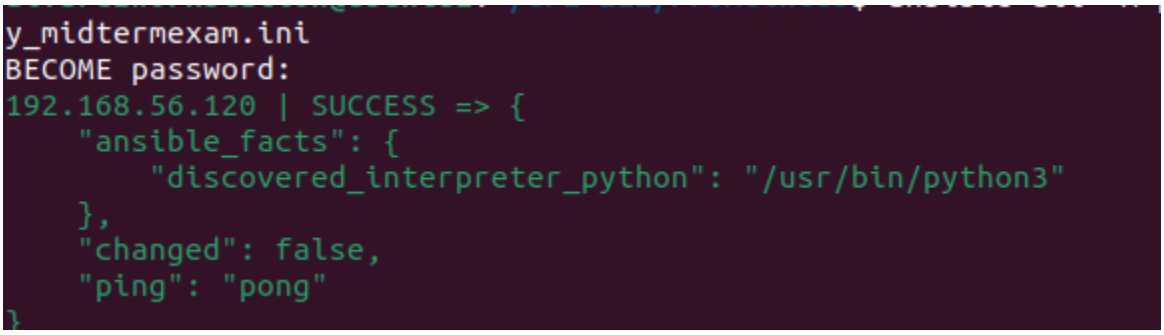
1. PROCEDURE OUTPUT

Sample output (students are allowed to add more steps as needed.)

Task 1: Prepare the Server



Server 1:



CentOS:

```
192.168.56.122 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
```

Updating packages:

```
---
- hosts: all
  become: true
  pre_tasks:
    - name: update repository index (CentOS)
      ansible.builtin.dnf:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      ansible.builtin.apt:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "Ubuntu"
```

```
TASK [1 (CentOS)] ***************
skipping: [192.168.56.120]
ok: [192.168.56.122]

TASK [1 (Ubuntu)] ***************
skipping: [192.168.56.122]
ok: [192.168.56.120]
```

Analysis: For this task, all I did was modify my inventory.ini for 2 nodes only then created a playbook that updates and upgrades packages for both nodes.

Task 2: Create the Prometheus User

```
tasks:
  - name: Creating Prometheus Group
    ansible.builtin.group:
      name: prometheus
      state: present
```

```
    - name: Creating Prometheus User
      ansible.builtin.user:
        name: prometheus
        group: prometheus
        shell: /usr/sbin/nologin
        system: true
        create_home: false
        state: present
```

```
TASK [2 Creating Prometheus Group] ******
ok: [192.168.56.120]
ok: [192.168.56.122]

TASK [2 Creating Prometheus User] *******
ok: [192.168.56.120]
ok: [192.168.56.122]
```

Analysis: In here, I separated the creating groups and creating users because it's easier to understand and modify if I did something wrong. I used shell: /usr/bin/nologin to block any interactive logins with message that displays "You cannot enter" or other ways to deny user accounts.

Task 3: Create Prometheus Directories

```
  - name: Create prometheus-directory
    ansible.builtin.file:
      path: "{{ item }}"
      state: directory
      owner: prometheus
      group: prometheus
      mode: '0755'
    loop:
      - etc/prometheus
      - var/lib/prometheus
```

```
TASK [3 Create Prometheus Directory] ****************************
ok: [192.168.56.120] => (item=etc/prometheus)
ok: [192.168.56.122] => (item=etc/prometheus)
ok: [192.168.56.120] => (item=var/lib/prometheus)
ok: [192.168.56.122] => (item=var/lib/prometheus)
```

Analysis: So here I use ansible.builtn.file to create directories and I use the {{ item }} to create two directories based on the loop section.

Task 4: Download Prometheus

```
    - name: Installing Prometheus
      ansible.builtin.unarchive:
        src: "https://github.com/prometheus/prometheus/releases/download/v3.14.>
        dest: /tmp
        remote_src: true
```

```
TASK [4 Installing Prometheus] ******
ok: [192.168.56.122]
ok: [192.168.56.120]
```

Analysis:
Here, I just copied my previous activity “Installing Nagios” and replaced the src and dest.
Task 5: Install the Prometheus Binaries

```
    - name: 5 installing the Prometheus Binaries
      ansible.builtin.copy:
        src: "/tmp/prometheus-3.14.0.linux-386/{{ item }}"
        dest: "/usr/local/bin/{{ item }}"
        owner: prometheus
        group: prometheus
        mode: '0755'
        remote_src: yes
      loop:
        - prometheus
        - promtool
```

```
TASK [5 installing the Prometheus Binaries] *************
changed: [192.168.56.122] => (item=prometheus)
changed: [192.168.56.122] => (item=promtool)
```

Analysis: In here, I used builtin.copy because of my previous seatwork. I copied the binaries on /tmp then moved it to /usr/local/bin.

Task 6: Install the Prometheus Configuration Files

```
  - name: 6 Install the Prometheus Configuration File
    ansible.builtin.copy:
      src: "/tmp/prometheus-3.14.0.linux-386/"
      dest: etc/prometheus/prometheus.yml
      owner: prometheus
      group: prometheus
      mode: '0644'
      remote_src: yes
```

```
TASK [6 Install the Prometheus Configuration File] *****
changed: [192.168.56.122]
```

(only centOS worked)
Analysis: In this task, I just copied the command like task 5 but here I used 0644 and moved it to etc/prometheus. 0644 makes the owner only read and write, not the full access.

Task 7: Validate the Configuration

- CentOS:

- Ubuntu:

Task 8: Create the systemd Service
Task 9: Start and Enable Prometheus
Task 10: Verify Prometheus
Task 11: Access Prometheus Through a Browser

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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